

Militarised Interstate Dispute Predicted Probabilities Dataset*

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May 2026

Abstract

This paper introduces a novel public dataset providing *ex ante* predicted probabilities of Militarized Interstate Dispute (MID) onset at the dyadic and monadic levels for the period 1992–2012. While existing quantitative research on political violence typically relies on realized conflict outcomes as explanatory variables, theoretical frameworks emphasize the importance of anticipated conflict risk in shaping state behavior regarding arming, alliance formation, and concessions. To address this gap, I estimate four statistical learning classifiers (logistic regression, elastic net, random forest, and gradient boosting) using 91 predictors. The random forest model achieves the strongest discriminative performance on a held-out test set (AUC = 0.994, precision = 0.889), substantially outperforming parametric alternatives. The resulting dataset offers researchers a theoretically grounded measure of latent conflict propensity that can be incorporated into models of militarization, preventive war, nationalist sentiment, and related phenomena.

*Word count $\approx$ 1,800.

1 Introduction

A large body of quantitative work on political violence uses the onset of Militarised Interstate Disputes (MIDs) as a proxy for external threat. Most existing approaches regress or condition on the *realised outcome* $MID_{ij,t}$ to explain contemporaneous behaviour. This overlooks a theoretically central quantity: *anticipation*.

As Fearon (1995) shows, states bargain in the shadow of war; the probability they attach to future militarised confrontation shapes their decisions regarding arming, alliances, and concessions before any dispute arises. For instance, Nordhaus et al. (2012) use their own constructed *ex ante* probability of dispute. However, they rely on a small set of predictors ($p = 10$) and only logistic regressions. More generally, such information is imprecise or completely unavailable in widely used IR datasets, and its absence may bias or omit entire mechanisms in models of militarisation, preventive war, or the rise of nationalist sentiment (Hegre and Nygård, 2015).

This project addresses this gap by generating two public datasets: a dyadic dataset containing $\hat{p}_{ij,t+1}$, the predicted probability that the dyad (i, j) will experience a MID onset in year $t + 1$, and a monadic dataset that aggregates these probabilities to the country-year level (noted $\hat{p}_{i,t+1}$). Four statistical learning classifiers are evaluated on a held-out test set; the predictions from the best-performing model serve as the variable of interest.

2 Data

2.1 Sources and construction

The dataset is constructed exclusively from publicly available data accessible through the `peacesciencer` R library (Miller, 2022), ensuring full reproducibility. The outcome variable is MID onset, drawn from the Correlates of War (CoW) MID project. Military capability indicators (CINC, military expenditure, personnel) come from the National Material Capabilities dataset (NMC v6.0). Democracy is measured with V-Dem’s Electoral Democracy In-

dex (`v2x_polyarchy`), chosen for its continuous, theoretically grounded scale. Alliance information (defence pacts, neutrality agreements, non-aggression treaties) comes from the Alliance Treaty Obligations and Provisions dataset (ATOP). Leadership transitions and irregular removals are taken from the Archigos dataset. Strategic rivalries are also included in `peacesciencer`. Trade flows and GDP figures combine the CoW Trade dataset and World Bank GDP estimates (constant 2011 USD).

Dyad-year observations are generated with `create_dyadyears()` using the CoW state system, with directed dyads set to `FALSE`, ensuring that each dyad appears only once. The panel covers the period 1946–2012; however, the empirical analysis is restricted to the years 1992–2012 in order to minimise the extent of missing values. The gross dataset contains over 360,000 observations from more than 19,000 dyads. After listwise deletion of incomplete cases, the analytic sample retains approximately 200,000 observations from over 11,800 dyads. This reduction reflects the fact that `create_dyadyears()` generates every theoretically possible dyad, including micro-states. Post-deletion, the sample is modestly biased towards richer and more democratic countries, a limitation acknowledged in the conclusion.

2.2 Predictors

The final feature set contains 91 predictors organised into five substantive clusters: (i) *past conflict patterns*: lagged MID onsets ($t - 1$, $t - 2$, $t - 5$), lagged highest level of violence, a five-year MID count, and the number of consecutive peace years; (ii) *military capability*: min/max CINC, military expenditure, personnel, and industrial capacity across the dyad; (iii) *economic structure*: min/max GDP, per-capita GDP, population, trade dependence, and GDP ratio (a proxy for power asymmetry); (iv) *political institutions*: V-Dem democracy scores, a joint democracy indicator, leadership transitions, irregular leadership changes, and the number of simultaneous leaders; and (v) *dyadic context*: land contiguity, water contiguity, log capital distance, ATOP alliance types, shared IGO memberships, and strategic rivalry indicators (positional, spatial, ideological, interventionary). This set reflects the

typical variables of quantitative IR approaches (Russett et al., 1995; Hegre and Nygård, 2015).

2.3 Train/test split

The dataset is partitioned into a training set (90%) and a held-out test set (10%) using stratified random sampling on the outcome class, preserving the class imbalance (approximately 0.3% MID-onset cases) in both splits.

3 Models

Let $\mathbf{X}_{ij,t}$ denote the feature vector for dyad (i, j) in year t , and let $y_{ij,t+1} \in \{0, 1\}$ denote MID onset in the following year. The goal is to estimate $\hat{p}_{ij,t} = P(y_{ij,t+1} = 1 \mid \mathbf{X}_{ij,t})$. Four classifiers are considered.

Logistic Regression. The baseline model includes no interaction or polynomial terms, keeping this naive model more interpretable. Given the large number of observations (200,000+), no explicit regularisation is applied.

Elastic Net. To reduce overfitting in the high-dimensional feature space, the elastic net minimises a penalised log-likelihood. Both hyperparameters α and λ —where $\alpha \in [0, 1]$ interpolates between Ridge ($\alpha = 0$) and Lasso ($\alpha = 1$), and $\lambda > 0$ is the overall penalty strength—are selected by stratified 5-fold cross-validation maximising the ROC AUC over a regular grid ($\lambda \in [10^{-4}, 10^0]$, $\alpha \in \{0, 1\}$). To allow the model to capture non-linearities, the recipe expands the continuous predictors with quadratic (`step_poly`, degree 2) terms prior to fitting; binary covariates are left untransformed. All predictors are standardised within each fold to avoid leakage.

Random Forest. A random forest aggregates $B = 100$ classification trees grown on bootstrapped subsamples (15 bootstrap resamples, stratified). Hyperparameters `mtry` $\in \{3, \dots, 20\}$ and minimum node size `min_n` $\in \{5, \dots, 50\}$

are tuned over a random grid of 15 combinations. The fitted engine (**ranger**) also records mean decrease in Gini impurity for variable importance.

Gradient Boosting (XGBoost). Gradient boosting constructs an additive model sequentially. To limit computational burden, tuning is performed over 15 random hyperparameter combinations using 15 bootstrap resamples (stratified). Three hyperparameters are tuned: tree depth $d \in \{3, \dots, 8\}$, learning rate $\eta \in \{10^{-3}, \dots, 10^{-1}\}$, and minimum node size $\text{min_n} \in \{5, \dots, 50\}$. Early stopping is not used; instead, a fixed number of trees ($B = 100$) is grown to reduce runtime. AUC is used as the primary criterion for hyperparameter selection, as it is robust to class imbalance ($\approx 0.3\%$ MID onsets).

4 Results

4.1 Performance assessment

All models are evaluated on the held-out test set, allowing us to compare the predictions with the true values. This comparison is shown in the following confusion matrices (figure 1).

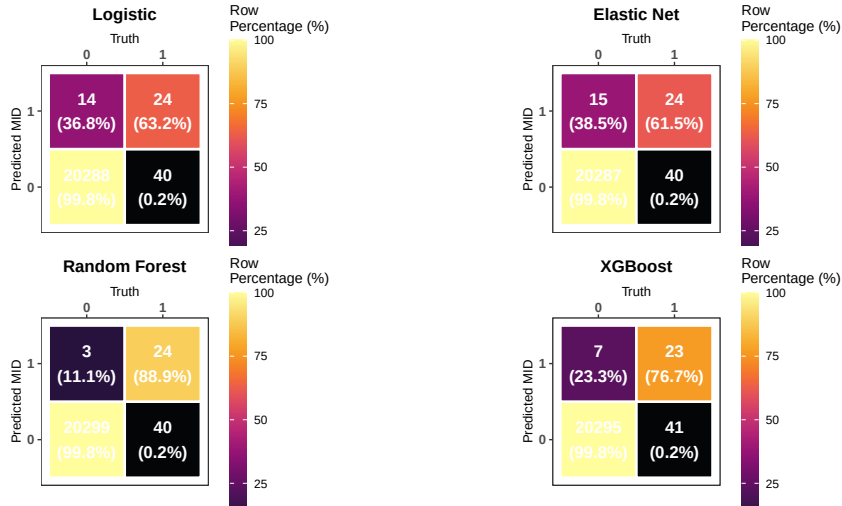


Figure 1: Confusion matrices

These results can be interpreted using four main metrics: (i) *accuracy* (share of correct classifications), (ii) *precision* (share of true MIDs among the predicted positives), (iii) *recall* (share of predicted MIDs among the true positives), and (iv) the *Area Under the ROC Curve* (which assesses discriminative power across all classification thresholds and is the criterion used to select optimal hyperparameters). All exact results are presented in table 1 of the Appendix alongside F1 scores.

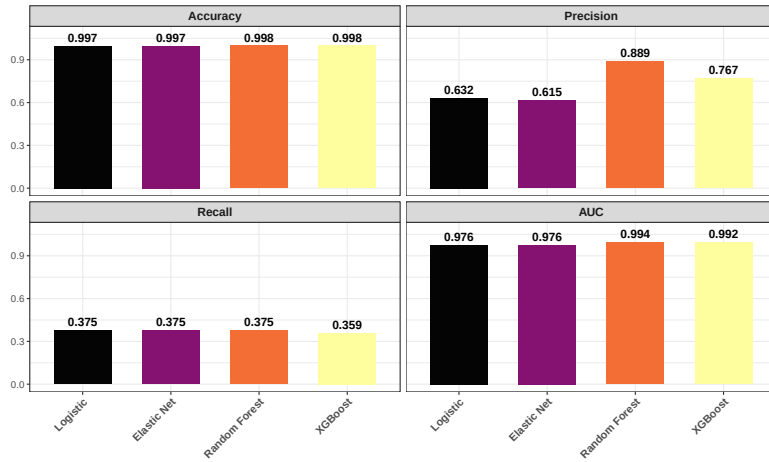


Figure 2: Model performance

Figure 2 shows the performance of all four models. All models achieve a very high level of accuracy (between 0.997 and 0.998). Nevertheless, this very high score could be misleading due to the extreme imbalance of the outcome variable ($E[Y_{ij,t+1}] = 0.003$). A trivial model that always predicts the majority class (no conflict) would achieve comparable accuracy without any genuine predictive power.

Precision represents the ability to correctly identify true positives among predicted positives. Here, the Random Forest demonstrates better discriminative ability, successfully distinguishing between predicted and realised MID events. Its precision (0.889) indicates that among 100 predicted disputes, roughly 89 actually occurred. This is 12 percentage points higher than the second-best performing model (XGBoost), and about 25 points higher than the worst models.

Recall, however, contrasts with the very high accuracy. The recall of the models is quite low (approximately 0.375, even for Random Forest). This means that out of 100 actual disputes, the models correctly identify only 36–37 of them.

Looking at the AUC (figure 3), the two tree-based ensemble methods substantially outperform the other classifiers: XGBoost achieves an AUC of 0.992, while Random Forest reaches 0.994 (compared to 0.976 for the other methods). Given that the Random Forest model consistently achieved the highest performance across all evaluated indicators, it was selected for the construction of the final datasets.

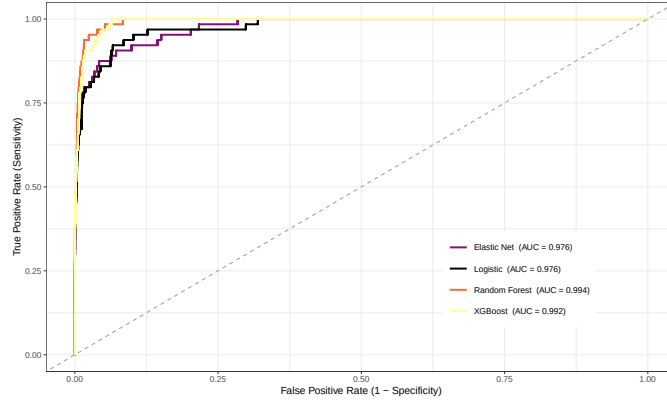


Figure 3: ROC curves

4.2 Top predictors

Figure 4 reports the variable importance from the Random Forest, measured by the mean decrease in the Gini impurity. Larger values indicate that splitting on a given predictor produces purer child nodes on average across all trees. The leading predictors are consistent with established quantitative IR findings: prior MID onsets, their duration and intensity are the strongest predictors of future onset, reflecting conflict recurrence and the persistence of underlying rivalries.

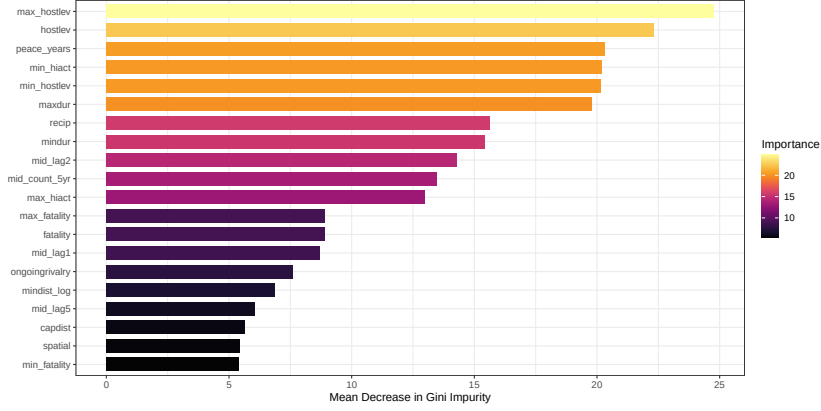


Figure 4: Top variable importances

5 Discussion and Conclusion

This project fills a gap in publicly available IR datasets by providing predicted probabilities of MID onset at $t + 1$. Four statistical learning models are estimated; the Random Forest achieves the best discrimination on the test set, with an AUC of approximately 0.994, a true positive rate (recall) of 0.375 at the default threshold, and a true negative rate of 0.998. Its predictions are then taken as the measure $\hat{p}_{ij,t+1}$ via `get_prob()`. A companion monadic dataset aggregates these to the country level using the complement rule, which gives the probability that country i experiences a MID onset with *at least one* partner, given by:

$$P(\text{MID}_{i,t+1}) = 1 - \prod_{j \neq i} (1 - \hat{p}_{ij,t+1}). \quad (1)$$

Equation (1) is equivalent to one minus the probability $\hat{p}$ that country i remains at peace with every possible partner j simultaneously. Then, both datasets are compiled and made available using:

```
library(readr)
monad_data <- read_rds(
  "https://github.com/SimonLabarthe/Militarised-Interstate-
  Dispute-Predicted-Probabilities-Dataset/blob/main/monad_data.rds"
)
```



```
dyad_data <- read_rds(
  "https://github.com/SimonLabarthe/Militarised-Interstate-
  Dispute-Predicted-Probabilities-Dataset/blob/main/dyad_data.rds"
)
```

Two limitations merit mention. First, complete-case selection introduces modest bias towards richer and more democratic countries; researchers using these predictions in analyses of conflict in the developing world should exercise caution. Second, the test sample performance (recall ≈ 0.375) implies meaningful irreducible uncertainty in predicting MIDs; the predicted probability should be treated as a noisy proxy for latent conflict propensity rather than a structural parameter. In this sense, a promising avenue for future research could be the inclusion of ideological parameters such as nationalism.

References

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Appendix

Supporting materials

Model	Accuracy	Precision	Recall	F1	AUC
Logistic	0.997	0.632	0.375	0.471	0.976
Elastic Net	0.997	0.615	0.375	0.466	0.976
Random Forest	0.998	0.889	0.375	0.527	0.994
XGBoost	0.998	0.767	0.359	0.489	0.992

Table 1: Model Performance Comparison